

Exploratory Data Analysis

GDP per Capita · Life Expectancy · Unemployment (WDI, 2022)

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2025-10-08

Agenda

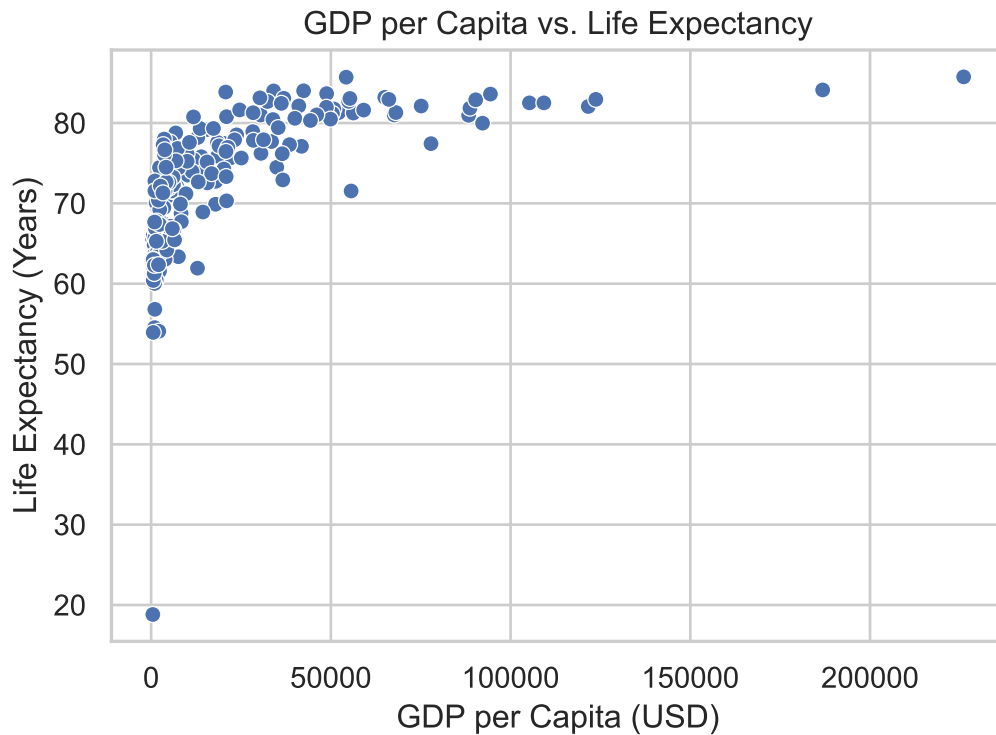
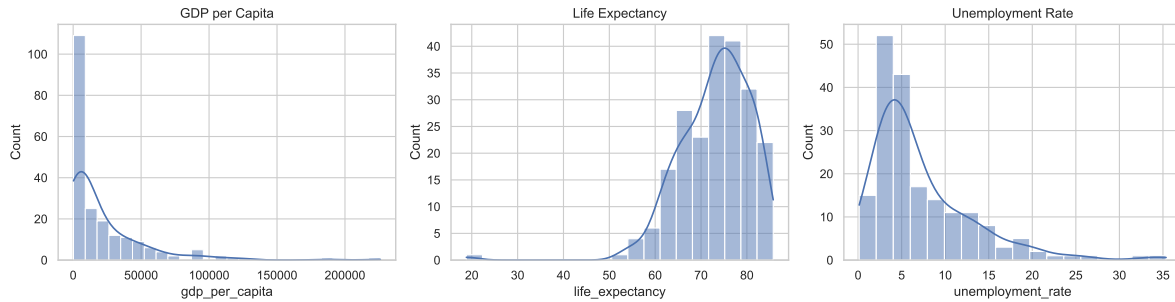
- Dataset snapshot
- Visual summaries
- U.S. vs. world
- Key takeaways

Data Overview

Three indicators were analyzed from the World Development Indicators (2022): 1. GDP per Capita (US\$) 2. Life Expectancy (years) 3. Unemployment Rate (%)

Distributions (2022)

- GDP per capita: The distribution is heavily right-skewed, showing that while a few high-income economies and developing regions, with small advanced economies (like Luxembourg or Singapore).
- Life expectancy: Most countries cluster between 65-80 years, reflecting global progress.
- Unemployment rate: The distribution is slightly right-skewed, with most nations maintaining rates below 10%. Outlier cases often represent economies facing structural challenges or recovery from shocks like the 2020 COVID-19 pandemic.

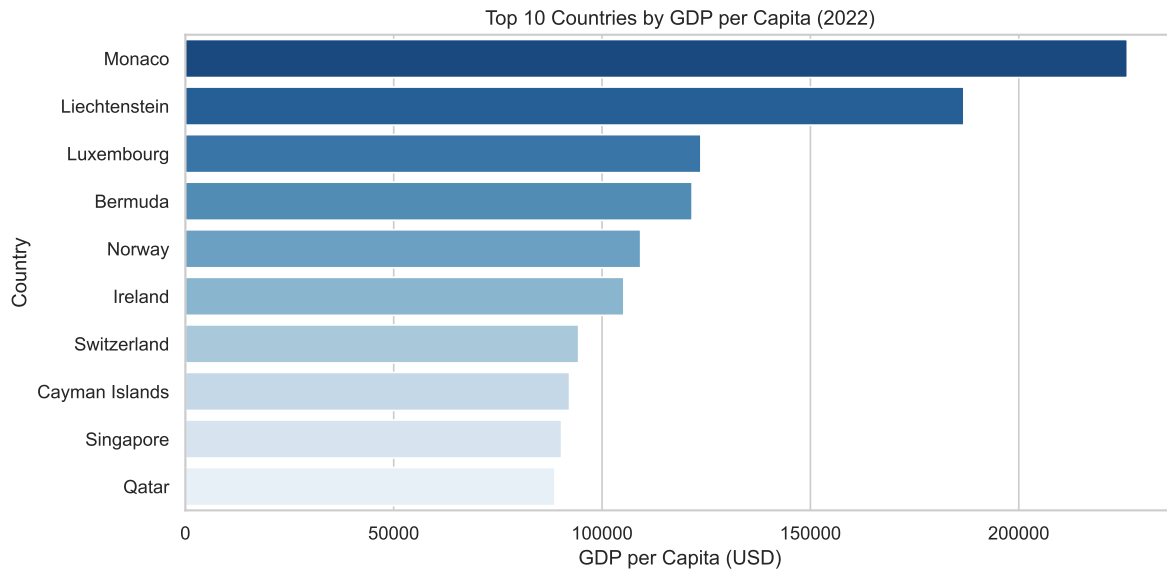


Top 10 by GDP per Capita

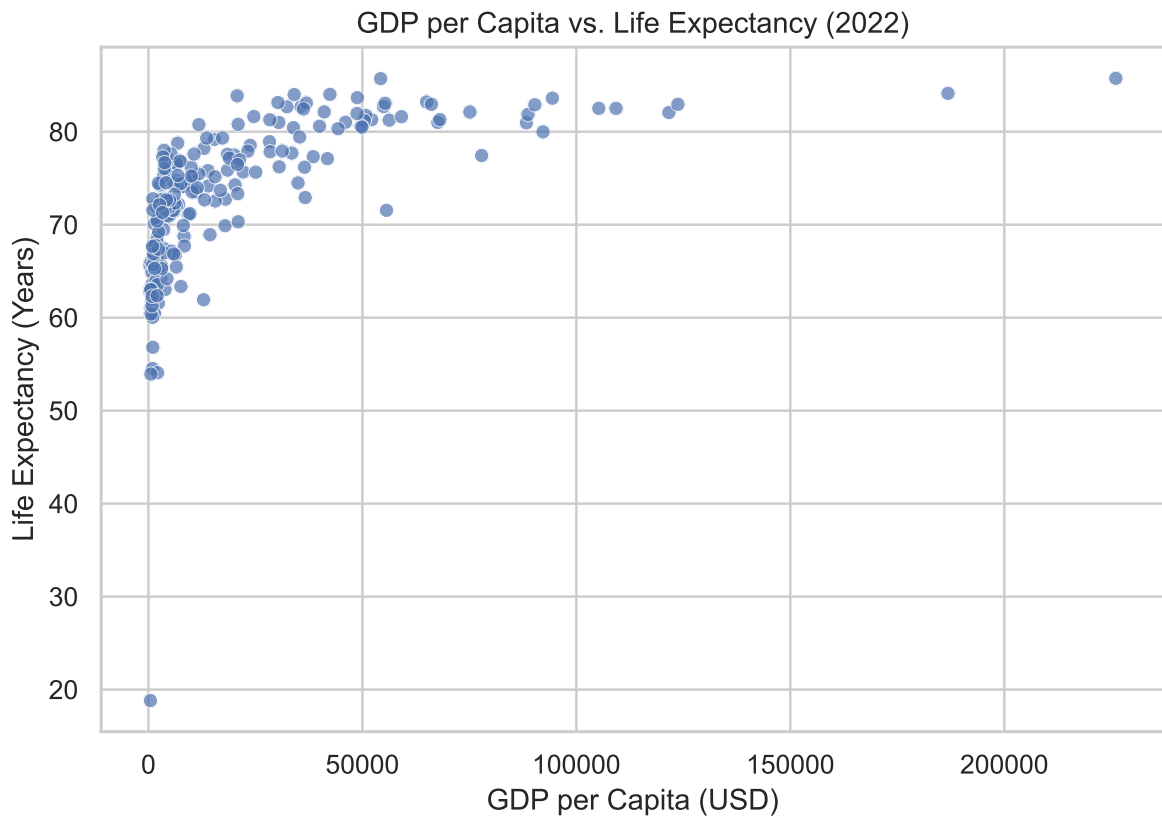
- Global income is heavily concentrated among a handful of advanced economies, where small, highly developed nations like Luxembourg and Switzerland far surpass the global average.
- This pattern underscores how wealth reflects economic specialization and productivity, not just population size.

GDP vs. Life Expectancy

- Higher income generally leads to longer life expectancy, reflecting how wealth improves access to healthcare, education, and living standards.



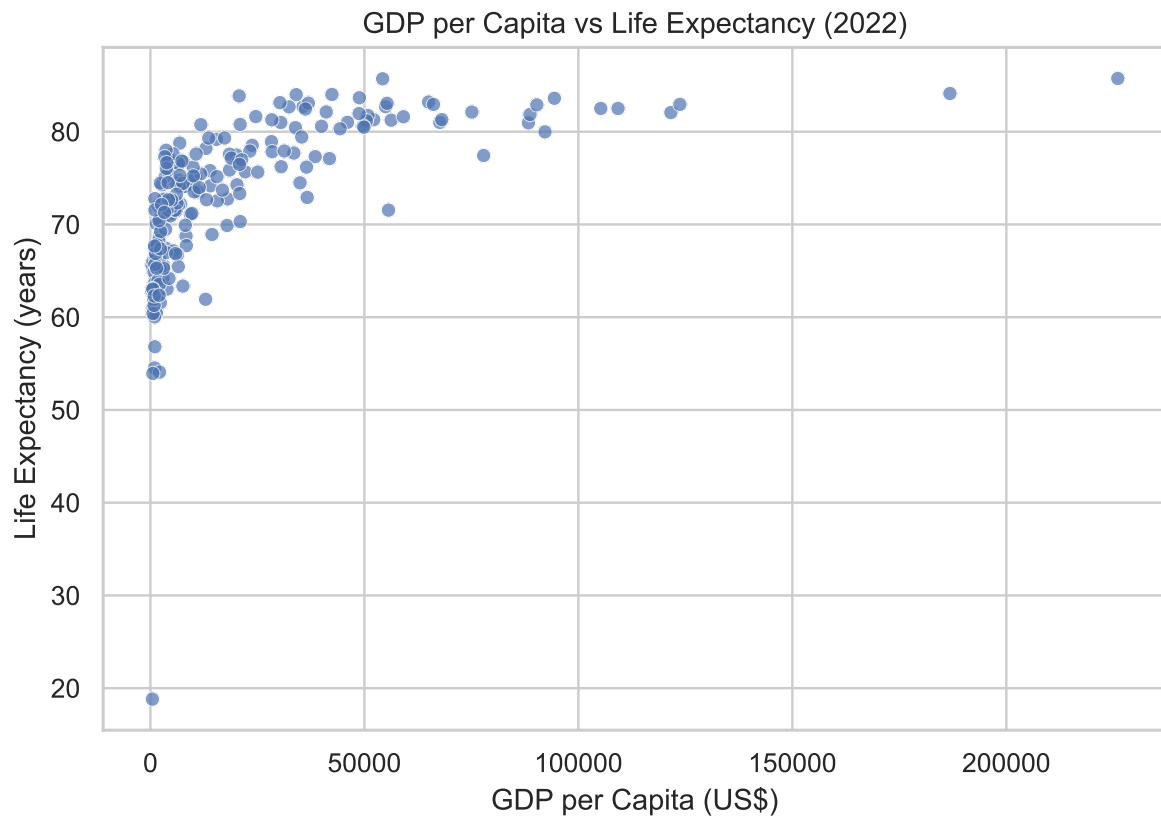
(a) Top 10 Countries by GDP per Capita (2022). Data source: World Bank.



(b)

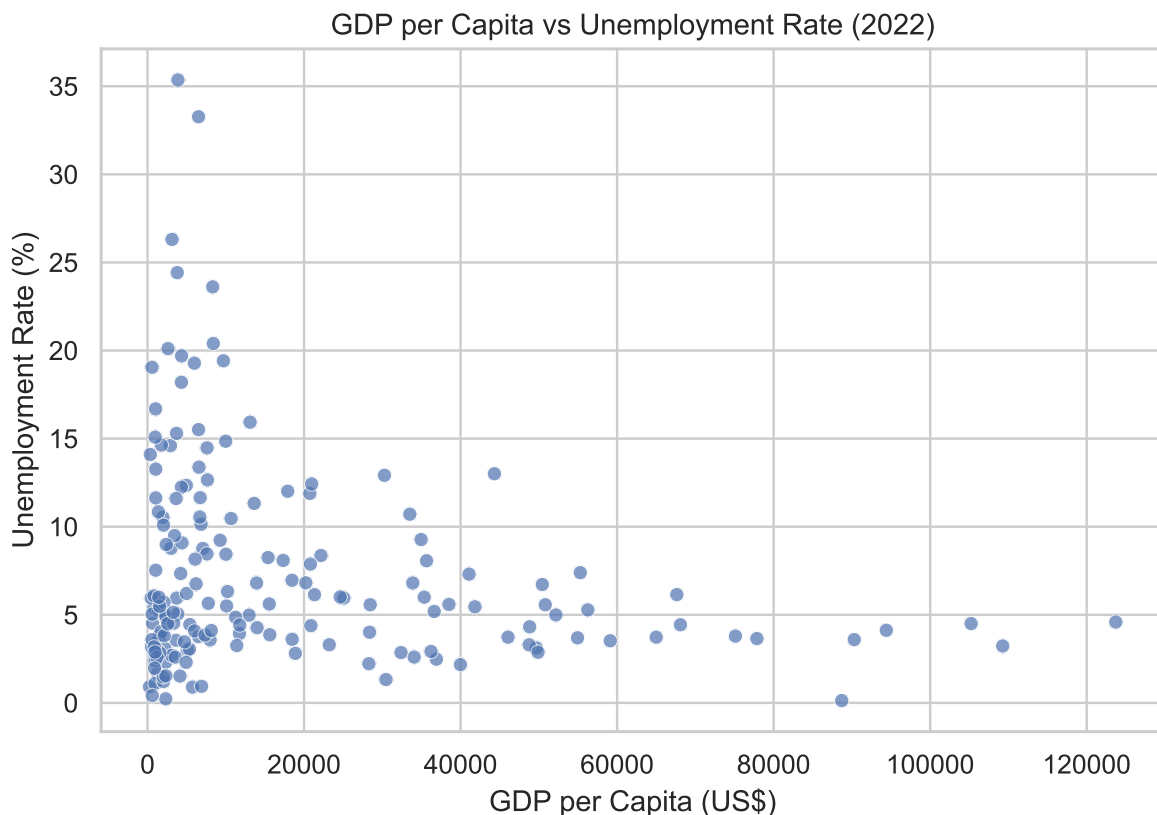
Figure 1

- However, the curve flattens at high income levels, showing diminishing returns once basic



GDP vs. Unemployment

- Wealthier nations generally experience lower unemployment rates, reflecting stronger economic growth and labor market conditions.
- Yet, the relationship is not uniform - unemployment also depends on policy choices, labor market structure, and demographic factors.



U.S. in Context

Summary • The United States combines extremely high income and low unemployment with one of the most productive economies in the world. • However, its life expectancy lags behind other high-income nations, highlighting inefficiencies in healthcare access, cost, and equity despite vast economic resources.

Table 1: United States vs. high-income and global averages (GDP per capita, life expectancy, unemployment rate) — World Bank WDI (2022).

Table 1

Indicator	United States	High-income average	Global average
GDP per capita (US\$)	77,860.91	69255.110000	21175.310000
Life expectancy (years)	77.43	81.280000	73.110000
Unemployment rate (%)	3.65	4.640000	7.230000

Conclusion

Overall, the analysis shows that higher national income is strongly associated with better health outcomes wealthier countries tend to enjoy longer life expectancy and greater economic stability. However, the benefits of income growth taper off among already affluent nations, where additional wealth brings smaller gains in longevity. Likewise, low unemployment alone does not guarantee stronger health outcomes, as broader social factors such as healthcare access, preventive care, and system efficiency play a far greater role. Ultimately, sustainable improvements in well-being depend not merely on economic expansion, but on how effectively countries convert their resources into equitable, accessible, and high-quality health systems.

References

Citations: • PwC Economic Outlook (Q1 2023) ((PwC) 2023) • WHO World Health Statistics 2023 (Organization 2023) • IMF World Economic Outlook 2024 (Fund 2024)

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